第4回 日中札幌計算機応用国際学術会議
THE 4th JAPANESE-SINO SAPPORO INTERNATIONAL CONFERENCE
ON COMPUTER APPLICATIONS

Morphological Analysis Method for Japanese Text Processing Using Machine Learning

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1.Introduction.

Japanese sentences are written as non-segmented character string without spaces between words. Therefore in the computer processing of Japanese sentences, such character strings must be divided into words. This dividing process is an important part of the morphological analysis in Japanese language processing [Koskenniemi, 1983] [Thurmair, 1984] [Russell et al., 1986] [Wothke, 1986]. A variety of analysis methods have been proposed in many research reports on this problem [Makino et al., 1980] [Abe et al., 1986] [Yamashina et al., 1988]. In these methods, words are recognized by collating each character pattern taken out of sentences with words stored in a dictionary.

In this paper, we propose a morphological analysis method based on machine learning. In this method, character patterns which exist commonly in several places in a text are extracted and registered in a dictionary as word candidates. Then several candidates which satisfy necessity conditions are defined as words.

The proposed method has following features.

1. The dictionary grows automatically by machine learning. It is not necessary to prepare a large number of words to be registered in the dictionary.
2. Contents of the dictionary can follow the change in word usage.
3. This method can determine a character string as a word, even if it is not registered in the dictionary.

2.Algorithm.

The proposed method consists of four processes as shown in Fig.1.

2.1 Inductive learning.

The basic elements of sentences are words. If a specific character string appears several times in a sentence, it is predictable to a word. Therefore, in the proposed method, character patterns appeared several times, i.e. common character patterns, in a text are extracted first. A common character pattern is called "Type I segment" and expressed as "S1".

Since a S1 may include plural words, the possibility of dividing it into words must be tried. To do this, "high order common character patterns" are extracted from S1s in the second step. A high order common character pattern is defined as a character pattern which appears in several S1s commonly. It is called "Type II segment" and expressed as "S2". The remainder of S1 after S2 extraction is called "Type III segment" and expressed as "S3". S1s, S2s and S3s are called "Word segment" and expressed as "WS". Fig.2 shows an example of S1 extraction. Fig.3 shows an example of WS extraction.

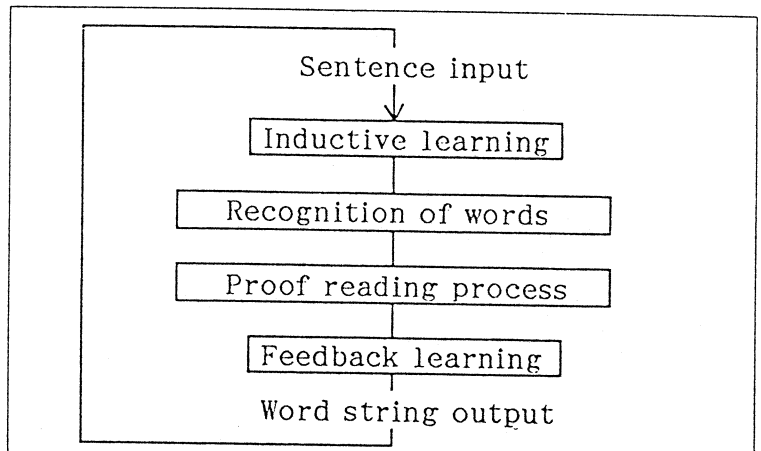


Fig.1 Word learning processes.

(1) Input sentences.
連続音声認識では連続音声を認識している。しかし、単音節音声認識では単音節ごとに認識する。

(2) Result of S1 extraction.
連続音声, 認識, 音声認識, では, 単音節

Fig.2 An example of S1 extraction.

(1) Result of S1 extraction (see Fig.4).
連続音声, 認識, 音声認識, では, 単音節

(2) Result of WS extraction.
(音声認識,認識)→認識:S2,音声:S3
(連続音声,音声)→音声:S2,連続:S3
では:S1
単音節:S1

(3) The WS extraction from input sentences.
連続(S3) 音声(S3) 認識(S2) では(S1) 連続(S3) 音声(S3) を 認識(S2) して
いる。しかし、単音節(S1) 音声(S3) 認識(S2) では(S1) 単音節(S1) ご
とに 認識(S2) する。

Fig.3 An example of WS extraction.

2.2 Recognition of words sentences.

In this step, words in input character string are recognized according to the certainty factors of each word candidates. Word candidates extracted from input sentences belong to one of following four kinds, namely, CW, S2, S1 and S3. Where, CWs are extracted in the feedback learning process described later. They are absolutely correct and are recognized as words first. They are registered in the dictionary whenever the feedback learning process is carried out, and initially there exist no CW in the dictionay. CWs have the highest certainty factor, and then S2s, S1s and S3s in order.

2.3 Proofreading process.

The results of the word recognition include several errors. Therefore in the next process, these errors are corrected manually. This process is called proofreading process.

2.4 Feedback learning.

The purpose of the feedback learning is to update the dictionary with the results obtained by the proofreading process.

The feedback learning is carried out as follows.

1. The recognition results are compared with those corrected in the proofreading process.
2. The words appeared in the recognition results and corrected in the proofreading process are registered as CWs in the dictionary.
3. The WSs identical with CWs are eliminated from the dictionary.
4. Likelihood values of correctly recognized words are increased, and those of erroneously recognized words are decreased.

A likelihood value (expressed by LV) of a word is calculated by the likelihood evaluation function (expressed by LEF) defined as

$LEF=\alpha*CF+AF-\beta*EF$:

where CF shows correctness frequency, AF shows appearance frequency, EF shows error frequency and α, β are coefficients, respectively.

3.Experiment for learning capability.

An experiment was carried out to assure the effectiveness of the proposed method. Four articles concerned with information processing were used for the experiment. Total number of characters of them was 35,867. In this experiment, the data were divided into 20 segments according to each chapter. The dictionary was empty at the beginning, and the learning process was repeated on each segment.

Fig.4 shows the change in the inclusion rate and the correctness rate. If we define,

A: the number of recognized words,
B: the number of correctly recognized words, and
C: the total number of words,

then the inclusion rate and the correctness rate are given as A/C and B/C, respectively.

The inclusion rates of the Total and the CW increase abruptly and reaches more than 95%. The inclusion rate of the WS decreases abruptly and falls less than 10%. The correctness rates of the Total and the CW

increase abruptly and reach more than 85% after input of about 18,000 characters to the system.

The rate of words which WSs dealed with is about 55% of the words that are not recognized by CWs. The rate recognized correctly by WSs is about 60% of the words that are not recognized by CWs. Therefore the rate recognized correctly by WSs is about 30% of the words that are not recognized by CWs.

The inclusion and correctness rates of WSs are high when those of CWs are low. This result shows that the words unrecognized by CWs are recognized by WSs to a certain degree.

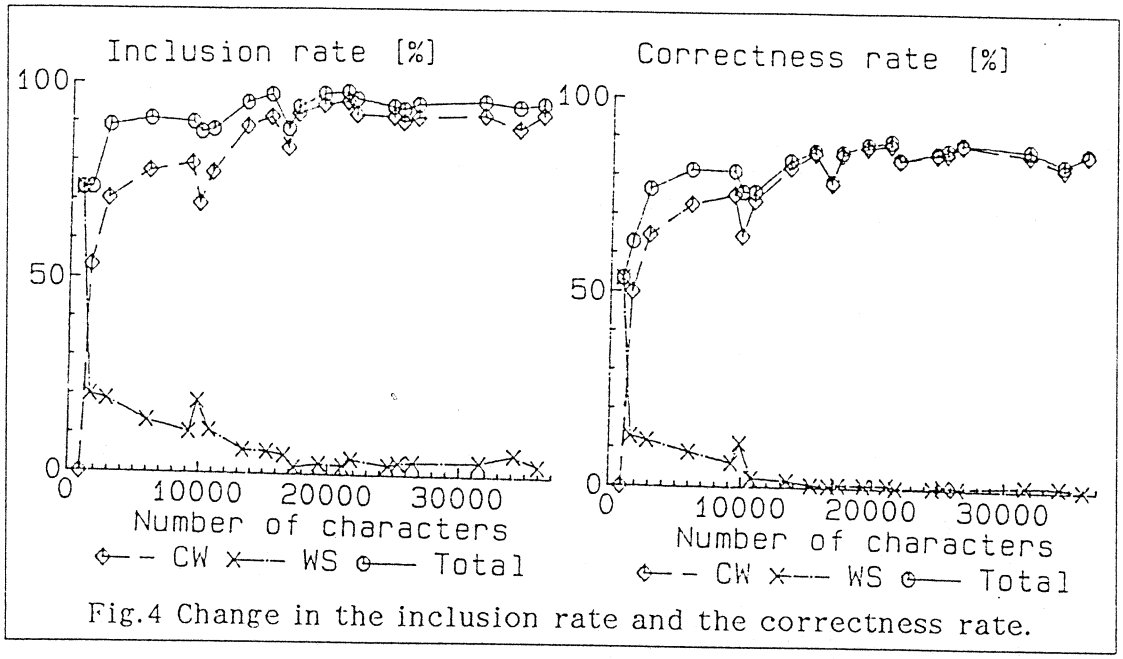


Fig.4 Change in the inclusion rate and the correctness rate.

4.Conclusion.

We proposed a morphological analysis method for Japanese text processing based on machine learning. Experimental results show that about 90% of words in a text are recognized correctly using the proposed method. It is shown that the proposed method is effective for the morphological analysis of Japanese sentences. As a future task, we will consider the learning capability of the proposed method to a variety of texts.

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